

BSDS (2024 Semester V): Regression Techniques

Midsem Part A

NAME :

ROLL :

Time: 60 minutes

Maximum attainable marks: 25

- I. Each following questions has 3 statements marked as A, B, and C. Indicate the correct statement(s) by placing a tick in the corresponding box. Note that, for a question all the options can be correct or incorrect. All the rough calculations related to a question should be done in the blank space below the question. $3 \times 6 = 18$

Q.1 Let $\underline{Y} = (Y_1, Y_2, Y_3)^\top$ be distributed as normal $N_3(\underline{\mu}, \Sigma)$, where $\underline{\mu} = (\mu_1, \mu_2, \mu_3)^\top$, and $\Sigma = ((\sigma_{i,j}))$.

- A. The conditional distribution of Y_1 given Y_2 will be $N(\mu_1, \sigma_{1,1})$ if and only if $\sigma_{1,2} = 0$. ☒
- B. Suppose $\underline{\mu} = \underline{0}$ and $\Sigma = UDU^\top$ where $D = \text{Diag}(d_1, d_2, d_3)$, $d_j > 0$ for all $j = 1, 2, 3$, and $UU^\top = U^\top U = I_3$. Then the components of $\underline{Z} = U^\top \underline{Y}$ are i.i.d.. ☒
- C. Consider the following choice of Σ : $\sigma_{j,j} = 1$ for $j = 1, 2, 3$, and $\sigma_{1,3} = \sigma_{2,3} = 0.5$. Then the conditional distribution of (Y_1, Y_2) given Y_3 has independent components. ☒

A. Marginal distⁿ of (Y_1, Y_2) is $N\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{pmatrix}\right)$.

Now, Y_1 and Y_2 are independent iff $\sigma_{12} = 0$

So, conditional distⁿ of Y_1 given Y_2 will be same as marginal, which is $N(\mu_1, \sigma_{11})$ if $\sigma_{12} = 0$.

B. $\underline{Z} = U^\top \underline{Y} \sim N(U^\top \underline{0}, U^\top \Sigma U) \equiv N(\underline{0}, D)$

$\therefore$ Marginal distⁿ of j th component of $\underline{Z}$ is : $Z_j \sim N(0, d_j)$
 $j = 1, 2, 3$

Unless, all the d_j s are same, the components of $\underline{z}$ are not iid.

$$C. \Sigma = \begin{pmatrix} 1 & \sigma_{12} & 0.5 \\ \sigma_{12} & 1 & 0.5 \\ 0.5 & 0.5 & 1 \end{pmatrix}$$

$$\therefore \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} \bigg| Y_3 = y_3 \sim N(\mu_{1 \cdot 2}, \Sigma_{11 \cdot 2})$$

$$\text{where } \Sigma_{11 \cdot 2} = \begin{pmatrix} 1 & \sigma_{12} \\ \sigma_{12} & 1 \end{pmatrix} - \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix} \begin{pmatrix} 0.5 & 0.5 \end{pmatrix} = \begin{pmatrix} 0.75 & \sigma_{12} - 0.25 \\ \sigma_{12} - 0.25 & 0.75 \end{pmatrix}$$

Unless, $\sigma_{12} = 0.25$, the off-diagonal elements of $\Sigma_{11 \cdot 2}$ are non-zero, which means conditional distⁿ of (Y_1, Y_2) given Y_3 does not have independent components.

Q.2 Consider the normal linear regression model $\underline{Y} = X\underline{\beta} + \underline{\varepsilon}$, where $\underline{Y} \in \mathbb{R}^n$, X is a $n \times p$ matrix of rank $\rho(X) = r < p$, and $\underline{\varepsilon} \sim N_n(0, \sigma^2 I)$, $\sigma^2 > 0$.

A. Define $RSS = \underline{Y}^T (I - P(X)) \underline{Y}$. Then $(n - p)^{-1} RSS$ is unbiased for σ^2 . ☒

B. The regression coefficients can not be estimated uniquely by the OLS method in this case. ☑

C. Define $\hat{\underline{\beta}} = (X^T X)^+ X^T \underline{Y}$. Then $\hat{\underline{\beta}}$ is independent of RSS . ☑

$$A. \quad RSS = \underline{Y}^T (I - P(X)) \underline{Y} = (\underline{X}\underline{\beta} + \underline{\varepsilon})^T (I - P(X)) (\underline{X}\underline{\beta} + \underline{\varepsilon}) = \underline{\varepsilon}^T (I - P(X)) \underline{\varepsilon}.$$

$$\underline{\varepsilon} \sim N(0, \sigma^2 I)$$

$$[\because P(X) \cdot X = X]$$

Both $P(X)$ and $I - P(X)$ are symmetric and idempotent, so $\text{rank}[I - P(X)] = \text{tr}[I - P(X)]$

$$= n - \text{tr}[P(X)]$$

$\therefore$ By Cochran's theorem, $\frac{RSS}{\sigma^2} \sim \chi^2_{n-r}$

$$= n - \text{rank}[P(X)]$$

$\therefore E\left[\frac{RSS}{\sigma^2}\right] = n - r \Rightarrow \frac{RSS}{n - r}$ is unbiased for σ^2 .

$$= n - \text{rank}(X)$$

$$= n - r$$

B. In this case, the least squares equation is, $X(X^T X)^+ X^T \underline{Y} = X\underline{\beta} \quad (*)$

As $\rho(X) = r$, there are r independent equations with $p (> r)$ unknowns.

As $X(X^T X)^+ X^T \underline{Y} \in \mathcal{C}(X)$, there are infinitely many solutions to $(*)$.

$$C. \quad \hat{\underline{\beta}} = (X^T X)^+ X^T \underline{Y} = (X^T X)^+ X^T X \underline{\beta} + (X^T X)^+ X^T \underline{\varepsilon} = \underline{\beta} + (X^T X)^+ X^T \underline{\varepsilon}$$

$$RSS = \underline{\varepsilon}^T (I - P(X)) \underline{\varepsilon} = \|(I - P(X)) \underline{\varepsilon}\|^2$$

$$\begin{aligned} \text{cov}(\hat{\underline{\beta}}, (I - P(X)) \underline{\varepsilon}) &= \text{cov}\left((X^T X)^+ X^T \underline{\varepsilon}, (I - P(X)) \underline{\varepsilon}\right) = \sigma^2 (X^T X)^+ X^T (I - P(X)) \\ &= \sigma^2 (X^T X)^+ (X^T - X^T P(X)) \\ &= \sigma^2 (X^T X)^+ (X^T - X^T) \\ &= 0. \end{aligned}$$

$\therefore \hat{\underline{\beta}}$ and $(I - P(X)) \underline{\varepsilon}$ are independent

$\therefore \hat{\underline{\beta}}$ and RSS are independent.

Q.3 Consider a set of p covariates $\{x_1, \dots, x_p\}$, a response y , and two sets of linear regression equations with n data points on the *centered* covariates:

$$\underline{Y} = \alpha_0 + \gamma_j \underline{x}_j + \underline{\varepsilon}_j, \quad \underline{\varepsilon}_j \sim N(0, \sigma_j^2 I), \quad j = 1, \dots, p, \quad (1)$$

and

$$\underline{Y} = \alpha_0 + \sum_{j=1}^p \beta_j \underline{x}_j + \underline{\varepsilon}, \quad \underline{\varepsilon} \sim N(0, \sigma^2 I), \quad (2)$$

where $\underline{Y} = (Y_1, \dots, Y_n)^\top$ is the vector of n samples on y and $\underline{x}_j = (x_{j,1}, \dots, x_{j,n})^\top$ is the vector of n centered observations on x_j , $j = 1, \dots, p$.

A. The OLS estimate of α_0 will be same in all the $p+1$ models. ☒

B. For any $j \in \{1, \dots, p\}$, the individual residual sum of squares, $RSS_j = \underline{Y}^\top (I - P(X_j)) \underline{Y}$ would not be smaller than the RSS obtained from the model in (2), where $X_j = [\underline{1}, \underline{x}_j]$ ☒

C. The OLS estimates of β_j and γ_j would have the same sign in models (1) and (2) for all $j = 1, \dots, p$. ☒

A. Model (1) can be written as, $\underline{Y} = (\underline{1} \quad \underline{x}_j) \underline{\gamma}_j + \underline{\varepsilon}_j$ where $\underline{\gamma}_j = \begin{pmatrix} \alpha_0 \\ \gamma_j \end{pmatrix}$

$$\begin{aligned} \text{Then } \hat{\underline{\gamma}}_{j, \text{OLS}} &= \begin{pmatrix} \underline{1}^\top \underline{1} & \underline{1}^\top \underline{x}_j \\ \underline{x}_j^\top \underline{1} & \underline{x}_j^\top \underline{x}_j \end{pmatrix}^{-1} \begin{pmatrix} \underline{1}^\top \underline{Y} \\ \underline{x}_j^\top \underline{Y} \end{pmatrix} \quad \underline{\varepsilon}_j \sim N(0, \sigma_j^2 I) \\ &= \begin{pmatrix} 1/n & 0 \\ 0 & 1/\|\underline{x}_j\|^2 \end{pmatrix} \begin{pmatrix} n\bar{Y} \\ \sum_i x_{ji} Y_i \end{pmatrix} \quad [\because \text{covariates are centered}] \\ &= \begin{pmatrix} \bar{Y} \\ \sum_i x_{ji} Y_i / \|\underline{x}_j\|^2 \end{pmatrix}, \quad j = 1, 2, \dots, p \end{aligned}$$

So, in all the p models in (1), $\hat{\alpha}_{0, \text{OLS}} = \bar{Y}$

Model (2) can be written as $\underline{Y} = X \underline{\beta} + \underline{\varepsilon}$ where $X = [\underline{1} \quad \underline{x}_1 \quad \dots \quad \underline{x}_p] = [\underline{1} \quad \underline{Z}]$

$$\begin{aligned} \text{Then } \hat{\underline{\beta}}_{\text{OLS}} &= (X^\top X)^{-1} X^\top \underline{Y} = \begin{pmatrix} 1/n & 0 \\ 0 & (\underline{Z}^\top \underline{Z})^{-1} \end{pmatrix} \begin{pmatrix} n\bar{Y} \\ \underline{Z}^\top \underline{Y} \end{pmatrix} \quad \underline{\beta} = (\alpha_0 \quad \beta_1 \quad \dots \quad \beta_p)^\top \\ &\quad \underline{\varepsilon} \sim N(0, \sigma^2 I) \end{aligned}$$

$$\therefore \hat{\beta}_{OLS} = \begin{pmatrix} \bar{Y} \\ (Z^T Z)^{-1} Z^T Y \end{pmatrix}$$

$$\therefore \hat{\alpha}_{0,OLS} = \bar{Y}$$

$\therefore$ OLS estimate of α_0 is same in all the $p+1$ models.

B. In model (2), $RSS = Y^T (I - P(X)) Y$

In model (1), $RSS_j = Y^T (I - P(\{x_j\})) Y$; $P(\{x_j\})$ is projection matrix for $[1 \ x_j]$

Both 1 and $x_j \in \mathcal{C}(X)$

$\therefore P(\{x_j\}) \cdot P(X) = P(\{x_j\}) \Rightarrow P(X) - P(\{x_j\}) = W$ satisfies $W^2 = W$.

$\therefore RSS_j - RSS = Y^T (P(X) - P(\{x_j\})) Y = Y^T W Y = \|WY\|^2 \geq 0$.

II. Give short answers to the following questions:

$3 \times 4 = 12$

Q.1 Let $\underline{Y} \sim N_p(\underline{0}, \Sigma)$ where $\Sigma > 0$. Define $\underline{Z} = A\underline{Y}$ and $\underline{W} = B\underline{Y}$, where A and B are $m \times p$ and $n \times p$ matrices, respectively. Find a sufficient condition on (A, B) to ensure that $\underline{Z}$ and $\underline{W}$ are independently distributed.

Let $\underline{V} = \begin{pmatrix} \underline{Z} \\ \underline{W} \end{pmatrix} = \begin{pmatrix} A \\ B \end{pmatrix} \underline{Y} = C\underline{Y}$, where C is a $(m+n) \times p$ matrix.

$$\therefore \underline{V} \sim N(\underline{0}, C\Sigma C^T)$$

Thus $(\underline{Z}, \underline{W})$ jointly follow a normal distⁿ.

Now, $\underline{Z}$ and $\underline{W}$ are independent iff $\text{cov}(\underline{Z}, \underline{W}) = 0$

$$\Leftrightarrow \text{cov}(A\underline{Y}, B\underline{Y}) = 0$$

$$\Leftrightarrow A\Sigma B^T = 0$$

So, $A\Sigma B^T = 0$ is the required condition.

Q.2 In a normal linear regression setup with covariates $\{x_1, \dots, x_p\}$, suppose we want to test if the effect of x_j on Y is two times stronger than that of x_{j-1} , $j = 2, \dots, p$. Express the hypothesis in the form $A\beta = c$, and write log-likelihood of the relevant parameters under the hypothesis.

$$H_0: \beta_j = 2\beta_{j-1} \text{ for } j = 2, \dots, p$$

$$S_0, \beta_2 = 2\beta_1, \beta_3 = 2\beta_2, \dots, \beta_p = 2\beta_{p-1}$$

$$\Rightarrow \underset{\sim}{A} \underset{\sim}{\beta} = \underset{\sim}{c} \quad \text{where } A = \underset{(p-1) \times (p+1)}{\begin{bmatrix} 0 & -2 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & -2 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & -2 & 1 \end{bmatrix}}; \quad \underset{\sim}{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}; \quad \underset{\sim}{c} = \underset{\sim}{0}$$

Now, under H_0 , $\beta_2 = 2\beta_1$, $\beta_3 = 2^2\beta_1$, ..., $\beta_p = 2^{p-1}\beta_1$

Hence, $\beta_{H_0} = (\beta_0, \beta_1 (2^0, 2^1, \dots, 2^{p-1}))^T$

So the log likelihood under H_0 is

$$\begin{aligned}
 & -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \|\underline{y} - \underline{X} \underline{\beta}_{H_0}\|^2 \\
 & = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^n \left\{ y_i - \beta_0 - \beta_1 \sum_{j=1}^p 2^{j-1} x_{ij} \right\}^2
 \end{aligned}$$

Q.3 Consider the normal linear regression setup, with $p = 5$ predictors and $n = 25$ observations. Suppose we are interested in finding a 95%-simultaneous confidence intervals (SCIs) for two particular linear combinations $\underline{a}_1^\top \underline{\beta}$ and $\underline{a}_2^\top \underline{\beta}$. Given the following tabulated values of F and t distributions, justify which of the following statements are correct:

- ✓ A. The Scheffe's interval for both the parametric functions are wider than Bonferroni's.
- B. The Scheffe's interval for both the parametric functions are narrower than Bonferroni's.
- C. The Scheffe's interval for one of the parametric functions is wider than Bonferroni's, and the other is narrower than that of Bonferroni's.

Distribution	F	F	F	F	t	t	t	t
df	(2, 20)	(2, 19)	(2, 20)	(2, 19)	20	20	19	19
α	0.05	0.05	0.025	0.025	0.025	0.0125	0.025	0.0125
Upper- α point	3.49	3.52	4.46	4.50	2.08	2.42	2.09	2.43

$$n - p - 1 = 19.$$

100(1- α)% SCI by Scheffe's method is :

$$\underline{a}_i^\top \underline{\beta} \in \underline{a}_i^\top \hat{\underline{\beta}}_{ols} \pm \sqrt{2 \frac{RSS}{19} F_{\alpha; 2, 19} \underline{a}_i^\top (\mathbf{X}^\top \mathbf{X})^{-1} \underline{a}_i}, i=1, 2, \forall \underline{a}_i \neq \underline{0}$$

100(1- α)% simultaneous CI by Bonferroni's method is :

$$\underline{a}_i^\top \underline{\beta} \in \underline{a}_i^\top \hat{\underline{\beta}}_{ols} \pm \sqrt{\frac{RSS}{19} t_{\frac{\alpha}{4}; 19}^2 \underline{a}_i^\top (\mathbf{X}^\top \mathbf{X})^{-1} \underline{a}_i}, i=1, 2$$

Bonferroni's interval is narrower than Scheffe's interval if

$$2 F_{\alpha; 2, 19} > t_{\frac{\alpha}{4}; 19}^2$$

$$\text{For } \alpha = 0.05, t_{\frac{\alpha}{4}; 19}^2 = t_{0.0125; 19}^2 = (2.43)^2 = 5.905$$

$$2 F_{\alpha; 2, 19} = 2 F_{0.05; 2, 19} = 7.04$$

$\therefore$ Scheffe's interval is wider for both parametric functions.

Rough work

Rough work